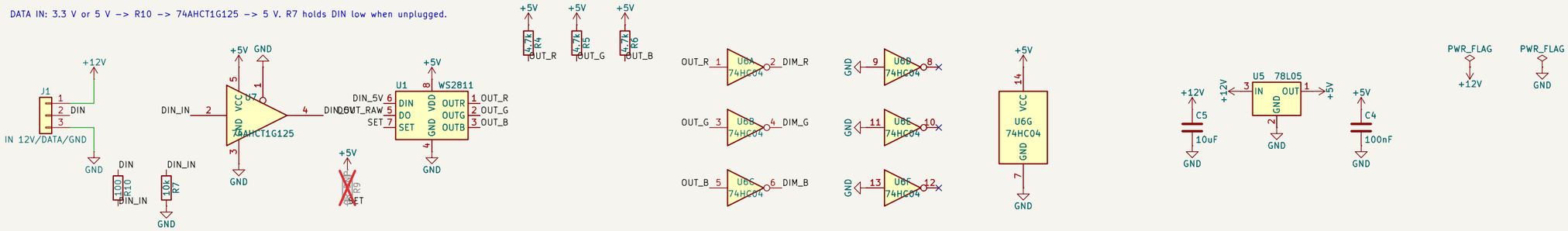
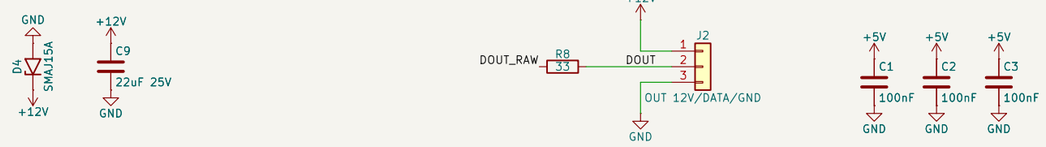
WS2811 outputs sink when ON; R4–R6 pull up; 74HC04 inverts so DIM is HIGH = LED on.

DATA IN: 3.3 V or 5 V -> R10 -> 74AHCT1G125 -> 5 V. R7 holds DIN low when unplugged.



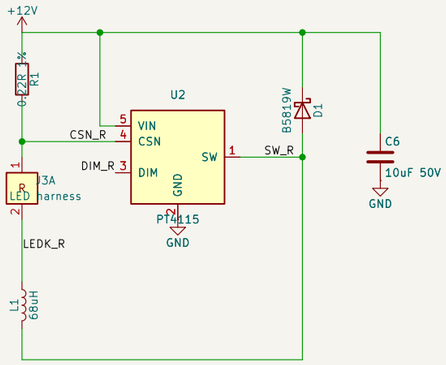


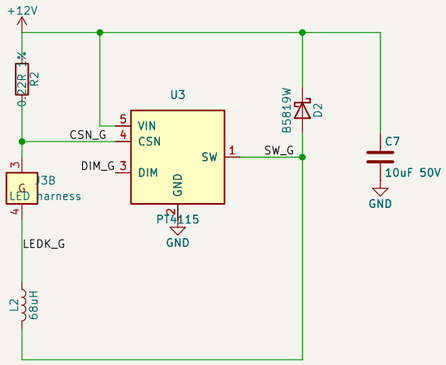
12 V in: SMAJ15A TVS + 22 uF for hot-plugged chain cables. J1/J2 12V and GND pass straight through (3 A).

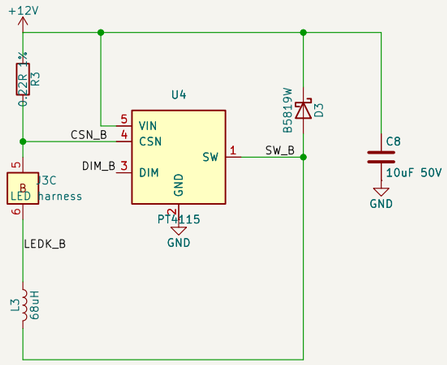
J1 and J2 (JST XH, back of the board) share one pinout: 1 = 12V, 2 = data, 3 = GND.

PT4115 loop (datasheet): +12V -> Rs -> J3 x+ -> LED -> J3 x- -> L -> SW; Schottky SW -> +12V. I_LED = 0.1 V / Rs = 455 mA

J3 = LED harness (JST PH 6, front of the board): 1 R+ 2 R- 3 G+ 4 G- 5 B+ 6 B-. Twist each +/- pair.







Generated by scripts/sch_driver.py
WS2811 pixel, 3x PT4115 455 mA, 12 V chain; LEDs on the aluminium LED board

Sheet: /
File: driver.kicad_sch

Title: RGB Spotlight – driver board

Size: A3	Date: 2026-10-05	**Rev: 2.0**
KiCad E.D.A. 10.0.6		Id: 1/1